SmartCivil : LLM with Construction Engineering Prior Knowledge using agentic graphRAG

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I. Introduction

Large language models (LLMs) have become strong general-purpose reasoners over natural language, yet when deployed as domain question-answering systems they exhibit a well-known failure mode: they hallucinate [3], producing fluent answers that are not supported by any authoritative source. In safety-critical engineering domains, where a single misquoted tolerance can invalidate a repair procedure, ungrounded generation is unacceptable. Retrieval-augmented generation (RAG) [2] mitigates this by conditioning the model on passages retrieved from a trusted corpus, so that generation is anchored to verifiable evidence rather than parametric memory.

The effectiveness of RAG is bounded by the quality of its retrieval stage. Sparse lexical retrieval such as BM25 [4] rewards exact term overlap and is robust for rare technical identifiers (for example, repair-method codes), whereas dense retrieval over learned embeddings [5] captures semantic similarity and paraphrase. Because the two signals are complementary, we adopt a hybrid retriever and combine its component rankings with reciprocal rank fusion (RRF) [6], a rank-based aggregation that requires no score calibration across retrievers. Beyond flat passage retrieval, a knowledge graph [7] imposes explicit structure on the corpus—entities such as defects, materials, and procedures, together with the relations among them—providing a grounding substrate that supports multi-hop and constraint-bearing queries that plain chunk retrieval handles poorly.

Two further technical challenges arise from the source material. First, the engineering knowledge exists only as scanned, image-based method statements, so the corpus must be reconstructed with optical character recognition (OCR) using a vision-language model [8] that can recover text, headings, and tabular structure. Second, the documents and the user population are bilingual (Thai and English) and frequently code-switched, which requires retrieval and generation to preserve technical terminology across languages. We further employ query decomposition for compound questions and constrain the generator to emit structured, citation-bearing answers so that every claim is traceable to a source page.

This paper makes the following technical contributions:

- a bilingual document-ingestion pipeline that converts scanned method statements into a structured, retrievable knowledge base via vision-language OCR;
- a hybrid keyword, full-text, and dense retriever whose rankings are fused by RRF and consumed by an LLM under a grounding constraint with page-level citation; and
- a multi-dimensional evaluation protocol—covering engineering-answer correctness, document grounding, bilingual fidelity, and latency—applied to a tunnel-segment repair corpus.

II. Methodology

Building on the RAG-LLM design of POLRAG [1], SmartCivil is organised as a six-phase pipeline that transforms scanned, image-based method statements into grounded, citation-bearing answers. The phases are document ingestion, knowledge-graph construction, query understanding, retrieval, answer generation, and structured formatting.

A. Document Ingestion

The source corpus consists of scanned bilingual (Thai–English) method statements for tunnel-segment repair. Each page is processed by a vision-language OCR model that recovers running text, headings, tables, and figures while preserving the page index. The output is serialised into a manifest together with extracted image and table assets, so that every text segment retains a back-pointer to its originating page; these page references are later surfaced as citations and are the basis of the document-grounding evaluation.

B. Knowledge-Graph Construction

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C. Query Understanding

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D. Retrieval

For each (sub-)question the system runs three retrievers in parallel. A keyword retriever scores each segment D against the query Q with Okapi BM25, which rewards shared terms while damping common words and long segments,

$$\text{BM25}(Q, D) = \sum_{q \in Q} \text{IDF}(q) \, \frac{tf(q, D)\,(k_1 + 1)}{tf(q, D) + k_1\left(1 - b + b\,\dfrac{|D|}{\text{avgdl}}\right)}, \tag{1}$$

where $tf(q, D)$ is the term frequency, $|D|$ the segment length, avgdl the mean segment length, IDF(q) the inverse document frequency, and $k_1 = 1.5$ and $b = 0.75$ control term-frequency saturation and length normalisation. A full-text retriever applies the same term-based scoring over both Thai and English surface forms. A dense retriever ranks segments by the cosine similarity between the question embedding $\mathbf{q}$ and each segment embedding $\mathbf{d}$,

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\|\,\|\mathbf{d}\|}. \tag{2}$$

The three rankings are merged by reciprocal rank fusion (RRF), which aggregates on rank position and therefore needs no score normalisation across retrievers,

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \tag{3}$$

where R is the set of retrievers, $\text{rank}_r(d)$ is the rank of segment d under retriever r, and k is a smoothing constant. The top-ranked fused segments form the context passed to the generator.

E. Answer Generation and Structured Formatting

A system prompt constrains the LLM to answer strictly from the retrieved context, to decline when the context is insufficient, and to respond in the language of the question. The raw answer is then passed through a formatting step that filters OCR noise and emits a structured response: the textual answer, any supporting image or table evidence, and the page-level document citation.

III. Experiment Design

We evaluate two complementary aspects of the system: the quality of the retrieval stage in isolation, and the end-to-end quality of the generated answers, both measured on a synthetic question–answer set.

A. Synthetic QA Dataset

The evaluation set is generated synthetically from the source repair manual by prompting an LLM to act as a domain QA-pair generator. The generator is constrained to use *only* information explicitly stated in the document—no external knowledge or general engineering practice—to include exact numeric values, to attach the originating page number to every answer, and to abstain when the evidence is unclear. Questions and answers are produced in the language of the source (Thai or English, frequently code-switched).

Each item carries a two-tier label. A *Rainbow Group* marks whether the question is Construction-Related (Group A) or Non-Construction (Group B, e.g. document or project metadata). Construction-Related items are further assigned a category—Repair Method, Material, Human Risk, or Environmental/Preventive Measures—each mapped to specific manual sections (Table I).

TABLE I
Rainbow-Group Category Taxonomy

Category	Source manual sections
Repair Method	Guideline for Repair; Cementitious Mortar; Epoxy Resin; Spalling Edge; Voids and Blow-holes
Material	Material Data Sheet
Human Risk	Risk Assessment; Material Safety Data
Environmental / Preventive	Preventative Measures and Environmental Impact
Non-Construction	Document/project metadata (Group B)

Two difficulty levels are evaluated. Each level merges a larger automatically generated set with a smaller hand-curated set (near-duplicate items removed): Level 1 comprises 113 questions (102 + 11) and Level 2 comprises 61 (44 + 17), for 174 in total. All results below are reported over these combined sets. The levels differ in reasoning demand:

- **Level 1 (single-category):** each question targets exactly one category and is answerable from a single section.
- **Level 2 (multi-category):** each question deliberately combines two or three categories and is multi-part, requiring evidence assembled from several sections—for example, selecting the correct repair method, then its material mixing ratio, then the associated environmental measure.

B. Retrieval Evaluation

Because segment-level relevance labels are unavailable, retrieval is assessed at the page level against the ground-truth page(s) attached to each question. Let $\mathcal{R}$ be the set of distinct pages among the top-k retrieved segments and $\mathcal{G}$ the ground-truth page set; the metrics are

$$\begin{aligned} &\text{Recall} = \frac{|\mathcal{R} \cap \mathcal{G}|}{|\mathcal{G}|}, \quad \text{Precision} = \frac{|\mathcal{R} \cap \mathcal{G}|}{|\mathcal{R}|}, \\ &\text{Hit@}k = \mathbb{1}[\mathcal{R} \cap \mathcal{G} \neq \varnothing]\,. \end{aligned} \tag{4}$$

Recall measures how completely the retriever surfaces the evidence pages, Hit@k whether at least one correct page appears in the top-k, and precision is bounded below because the system returns k pages where typically only one is relevant. These page-level scores serve as a practical proxy for context recall and precision.

C. LLM Performance

End-to-end answer quality is scored on a 100-point rubric spanning five dimensions, summarised in Table II. Engineering-answer correctness and document-grounding accuracy carry the largest weights because they capture, respectively, whether the prescribed repair method matches the method statement and whether the answer is faithful to the source documents without hallucination.

Each scored dimension is graded into qualitative bands. For engineering-answer correctness, an answer is *Excellent*

TABLE II
LLM Performance Evaluation Rubric

Evaluation Dimension	Max	Metric / Method
Engineering Answer Correctness	40	Expert validation, rule matching
Document Grounding Accuracy	30	Citation correctness, hallucination rate
Bilingual Capability (TH–EN)	10	Translation accuracy, semantic consistency
Response Latency	10	Average response time (s)
System Prompt Compliance	10	Prompt adherence score
Total	**100**	

(36–40) when it identifies the correct repair method in line with the method statement (e.g. RP-TN-01, RP-TN-02, or Reject) together with the complete numeric conditions such as crack width, depth, and void size; *Good* (26–35) when the method is correct but some numeric detail is missing; *Fair* (16–25) when the intent is partially correct but the chosen method departs from the guideline; and *Poor* (0–15) when the method is wrong or contradicts the documents. For document-grounding accuracy, an answer is *Excellent* (26–30) when it is entirely supported by the corpus with no hallucination, *Good* (18–25) when citations are largely correct with minor over-summarisation, and *Poor* (0–17) when it contains clear hallucination or unsupported claims. Response latency is mapped from the mean per-question time: ≤ 5 s scores 10, 6–10 s scores 8, 11–20 s scores 5, and > 20 s scores 0.

IV. Results and Discussion

We evaluate on two question sets: a single-category set (Level 1, $n = 113$) and a multi-category set that pairs two or three topics per question (Level 2, $n = 61$). The reported configuration uses multilingual dense embeddings for the semantic retriever and disables LLM query rewriting, a setting selected because an ablation on a held-out subset showed rewriting degraded retrieval (it lowered keyword Hit@k on the multi-category set from 0.75 to 0.55).

A. Retrieval

We report page-level recall, precision, and Hit@k defined in (4) for each retriever, first aggregated over the full set and then broken down by the document section (sub-category) each question targets. Precision is uniformly low because the system returns $k = 5$ pages per query while most answers reside on a single page, so recall and Hit@k are the informative signals.

1) Overall: On the full set (Level 1 $n = 113$, Level 2 $n = 61$, Table III) the dense (*vector*) retriever is the strongest method on both levels, and RRF fusion—dominated here by the weaker keyword ranking—trails it rather than improving on it, indicating that once dense retrieval is reliable the lexical signal adds little for this corpus. Dense retrieval is decisive only with genuine semantic embeddings: with a non-semantic baseline embedding the vector retriever collapsed (Level 2 Hit@k 0.40) and fusion added nothing, whereas with real embeddings it reaches 0.87.

TABLE III
Retrieval Quality by Method — Full Set (Page-Level)

Retriever	Level 1			Level 2		
	Rec.	Prec.	Hit	Rec.	Prec.	Hit
Keyword (BM25)	0.49	0.10	0.49	0.42	0.14	0.61
Vector (dense)	**0.81**	**0.18**	**0.81**	**0.64**	**0.22**	**0.87**
Fused (RRF)	0.76	0.17	0.76	0.57	0.19	0.74

2) By Document Section: Table IV breaks Level-1 Hit@k down by the document section (sub-category) each question targets. Vector retrieval is the strongest or tied method in almost every section. The keyword retriever all but fails on the Thai-prose-heavy Preventative Measures and Material Safety Data sections (Hit@k 0.08 and 0.00), where exact-term overlap is rare, yet leads on the identifier-dense Repair Note; this complementarity is what fusion exploits in a few sections (e.g. Material Approval). The Out-of-Scope (non-construction) questions correctly retrieve no relevant page. Level 2 is omitted here because its multi-section questions leave mostly singleton groups.

TABLE IV
Level-1 Retrieval Hit@k by Document Section

Document Section	n	KW	Vec	Fus
Material Data Sheet	33	0.67	**0.85**	**0.85**
Guideline for Repair	18	0.67	**0.89**	0.78
Preventative Measures	12	0.08	**1.00**	**1.00**
Material Safety Data	8	0.00	**0.88**	0.50
Cementitious Mortar	8	0.25	**0.50**	0.38
Epoxy Resin	8	0.50	**0.75**	**0.75**
Risk Assessment	7	0.57	**1.00**	**1.00**
Spalling Edge	5	0.40	**0.80**	0.60
Repair Note	4	**1.00**	0.50	**1.00**
Voids (High-Str. Mortar)	4	**0.50**	**0.50**	**0.50**
Material Approval	2	0.50	0.50	**1.00**
Out of Scope	2	0.00	0.00	0.00
Voids & Blowholes	2	0.50	**1.00**	0.50

B. End-to-End Answer Quality

Each fused-retrieval answer was scored against an expert ground truth on the rubric of Table II using an LLM-as-judge protocol over the full evaluation set; latency is derived from the measured per-question time, and bilingual capability and system-prompt compliance are assessed at the system level. To gauge the reliability of automatic grading, every answer was scored independently by two judges from different model families—a Claude model as the primary judge and a Gemini model as the secondary; Table V reports the primary (Claude) judge and Section IV-B2 the agreement between them.

1) Overall: Engineering-answer correctness is solid on both sets (31.8 and 28.4 of 40): when the right page is retrieved, the model identifies the correct repair method and quantitative conditions, and multi-category questions are usually answered across all their sub-parts. The lower Level-2

TABLE V
EVALUATION RESULTS (FUSED RETRIEVAL, FULL SET)

Dimension	Max	L1	L2
Engineering Answer Correctness	40	31.8	28.4
Document Grounding Accuracy	30	23.3	21.8
Response Latency	10	5.9	3.8
Bilingual Capability (TH–EN)	10	9.0	9.0
System Prompt Compliance	10	9.0	9.0
Total	**100**	**79.0**	**72.0**

score reflects its harder, compound questions and a higher rate of honest abstentions when the evidence is not retrieved, together with occasional omission of a second numeric sub-part (e.g. a mixing ratio or pot life).

Table VI contrasts a high- and a low-scoring answer (both translated from Thai). The Excellent answer reproduces the correct method- statement criteria with complete numeric bounds and a correct page citation; the Poor answer is fluent and well-formed but reports a numeric value that contradicts the data sheet, which is precisely the failure mode the correctness dimension is designed to penalise.

TABLE VI
EXAMPLE ANSWERS AT HIGH AND LOW CORRECTNESS

Excellent — 38/40	
Question	Repair criteria for a “Hard Crack” repaired with Cementitious Mortar?
Ground truth	width $0.1 < A < 0.3\,\mathrm{mm}$ and depth $> 15\,\mathrm{mm}$.
System	“width (A) $> 0.1\,\mathrm{mm}$ but $< 0.3\,\mathrm{mm}$; depth $> 15\,\mathrm{mm}$” — cites *Guideline for Repair*, p. 3.
Why	correct criteria, complete numeric bounds, correct citation.
Poor — 10/40	
Question	Bond strength to concrete of Sikadur-32TH?
Ground truth	2.5–3 $\mathrm{N/mm^2}$ (concrete failure).
System	“$> 4\,\mathrm{N/mm^2}$ for dry and moist concrete.”
Why	wrong numeric value; contradicts the data sheet despite fluent form.

2) Judge Agreement: LLM-as-judge scoring is itself a model output, so we cross-validate the primary judge (Claude) against a second judge from a different model family (Gemini) over all 174 answers (Table VII). Because the system's answers are themselves generated by a Gemini model, using a different-family model as the primary judge avoids same-family self-preference. The two judges agree closely on engineering correctness (Pearson $r = 0.93$, mean absolute difference 4.3 of 40) but diverge on grounding ($r = 0.86$, yet an 8.8-point gap): the Gemini judge is systematically stricter, treating the multi-page citations as a grounding failure rather than awarding partial credit when the correct page is among those returned. The strong rank correlations indicate the judges order answers consistently; the grounding offset is a calibration difference, so absolute grounding scores should be read with a margin.

TABLE VII
INTER-JUDGE AGREEMENT (ALL 174 ANSWERS)

Dimension	Claude	Gemini	r	$\|\Delta\|$
Eng. Correctness (/40)	30.6	28.5	0.93	4.3
Doc. Grounding (/30)	22.8	14.3	0.86	8.8

3) By Document Section: Table VIII breaks the two content dimensions of the rubric down by document section on the full Level-1 set. Correctness and grounding are highest on the well-structured Guideline, Risk Assessment, Preventative Measures, and Repair Note sections, and lowest on Material Safety Data and the sparse Voids questions—mirroring the retrieval difficulty in those sections, since an answer can only be correct and grounded when its evidence page is retrieved.

TABLE VIII
LEVEL-1 RUBRIC (CORRECTNESS, GROUNDING) BY DOCUMENT SECTION

Document Section	n	Eng/40	Grnd/30
Material Data Sheet	33	29.8	23.6
Guideline for Repair	18	36.6	23.6
Preventative Measures	12	**38.6**	**26.3**
Material Safety Data	8	22.2	20.1
Cementitious Mortar	8	30.8	19.5
Epoxy Resin	8	29.8	22.4
Risk Assessment	7	36.3	25.9
Spalling Edge	5	30.0	21.6
Repair Note	4	38.2	26.0
Voids (High-Str. Mortar)	4	21.0	21.0
Material Approval	2	38.0	26.0
Out of Scope	2	33.0	22.0
Voids & Blowholes	2	22.0	21.0

As a second, fully-automatic per-section view, we also report the answer-to-ground-truth similarity (AnsSim, the cosine similarity between the generated and reference answer embeddings), broken down by document section on the full Level-1 set (Table IX). The trend mirrors retrieval: the vector and fused configurations produce answers closer to the reference than keyword in almost every section, and the gap is widest on Preventative Measures ($0.63 \rightarrow 0.85$), exactly where keyword retrieval failed. AnsSim is only a coarse proxy—it rewards surface overlap rather than engineering correctness—so it is reported here to expose per-section trends, not as a substitute for the rubric of Table V.

C. Document Grounding and Citation

Grounding scores in the *Good* band (23.3 and 21.8 of 30): answers are overwhelmingly faithful to the corpus and carry chapter/page citations, but the score is capped by noisy citations—the system returns several candidate pages where typically one is correct, and a handful of otherwise-correct answers cite an adjacent page. Hallucination is rare. Table X illustrates the distinction from correctness: a *factually correct* answer can still score low on grounding when it cites the wrong source page.

TABLE IX
LEVEL-1 ANSWER SIMILARITY (ANSSIM) BY DOCUMENT SECTION

Document Section	n	**KW**	**Vec**	**Fus**
Material Data Sheet	33	0.71	**0.72**	**0.72**
Guideline for Repair	18	0.76	**0.79**	**0.79**
Preventative Measures	12	0.63	**0.85**	0.84
Material Safety Data	8	0.68	**0.78**	0.73
Cementitious Mortar	8	0.75	0.80	**0.81**
Epoxy Resin	8	0.74	**0.78**	0.77
Risk Assessment	7	0.72	0.77	**0.78**
Spalling Edge	5	0.79	**0.82**	0.81
Repair Note	4	0.74	0.66	**0.76**
Voids (High-Str. Mortar)	4	0.74	**0.75**	0.74
Material Approval	2	0.79	0.77	**0.82**
Out of Scope	2	**0.66**	**0.66**	0.65
Voids & Blowholes	2	0.76	**0.84**	0.74

TABLE X
EXAMPLE ANSWERS AT HIGH AND LOW GROUNDING

Excellent — 27/30	
Question	First aid if Sika Skim Coat enters the eye?
Ground truth	remove contact lenses, rinse with water $\geq$ 10 min, see a doctor (p. 125).
System	states the same three steps and cites *Materials Safety Data*, p. 125.
Why	fully supported by the corpus, correct page citation, no hallucination.
Poor — 15/30	
Question	Project name stated in the document?
Ground truth	Sewerage Tunnel Bung Nong Bon to Chaophraya River (p. 1).
System	names the correct project but cites p. 147 (a material-approval page).
Why	fact is right, but the cited page is unsupported (page-hit miss); grounding penalises the citation error.

D. Latency

Dropping the second answer-formatting pass roughly halves response time relative to the full pipeline, giving means of 11.9 s (Level 1) and 17.6 s (Level 2). Most questions fall in the 6–20 s bands, giving latency scores of 5.9 and 3.8 of 10; the multi-category set is slower because its compound questions retrieve and synthesise more context. Latency nonetheless remains the lowest-scoring dimension and the clearest target for further optimisation.

E. Bilingual Fidelity and Compliance

Bilingual capability and prompt compliance are strong: answers consistently match the language of the question, preserve English material names and repair-method identifiers word-for-word within Thai prose, cite their sources, and abstain when the context is insufficient. Table XI contrasts a high- and a low-fidelity answer. In the Excellent case a Thai question is answered in Thai while the English method code *RP-TN-02* and material names are preserved word-for-word, so a bilingual site team reads it without translation loss. The Poor case reproduces an OCR-corrupted Thai term word-for-word, leaving a safety hazard unintelligible; genuine wrong-language responses were not observed, so terminology corruption from noisy OCR is the dominant bilingual failure mode.

TABLE XI
EXAMPLE ANSWERS AT HIGH AND LOW BILINGUAL FIDELITY

Excellent — 9/10	
Question	รอยร้าวแบบ Hard Crack กว้าง 0.4 มม. ลึก 20 มม. ใช้วิธีซ่อมรหัสใด
System	ใช้วิธี RP-TN-02 (Repair with Epoxy Resin) เนื่องจากรอยร้าวกว้าง 0.3--0.5 มม. และลึกมากกว่า 15 มม.
Why	answered in Thai with the English method code and material names kept word-for-word; meaning fully intact.
Poor — 3/10	
Question	อันตรายจากการใช้เครื่องเจียรคอนกรีตมีอะไรบ้าง
System	ไฟฟ้าดูด และ โคนใบเจียรมาค — corrupted; should read โดนใบเจียรบาด (“cut by the grinder blade”).
Why	passes an OCR-garbled Thai term through word-for-word, so the second hazard is unintelligible.

The defining compliance behaviour is correctly *declining* an unanswerable question rather than guessing; Table XII contrasts this with the opposite failure, where the system over-generates beyond the retrieved context.

TABLE XII
EXAMPLE ANSWERS AT HIGH AND LOW PROMPT COMPLIANCE

Excellent — 10/10	
Question	Pot life of Sikadur-31 CF Normal at +30°C? (not in the corpus)
System	“the provided document does not contain this information.”
Why	obeys “answer only from context, decline when insufficient”; stays in scope, correct language and format.
Poor — 3/10	
Question	Project name stated in the document? (earlier configuration)
System	returns a list of $\sim$25 unrelated project names instead of the single relevant answer or a decline.
Why	out-of-scope over-generation; violates “answer strictly from the retrieved context.”

Overall the framework is reliable when retrieval succeeds; with semantic embeddings in place, latency and citation precision are the remaining priorities for future work.

V. CONCLUSION AND FUTURE WORK

[Incomplete — to be written.]

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REFERENCES

[1] H. Lin, P. Deng, Q. Zhong, and X. Zhu, “POLRAG: A RAG-LLM framework for policy question answering,” in *Proc. 2025 IEEE Int. Conf. High Performance Computing and Communications (HPCC)*, 2025, pp. 1259–1264, doi: 10.1109/HPCC67675.2025.00179.

[2] P. Lewis, E. Perez, A. Piktus, et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[3] Z. Ji, N. Lee, R. Frieske, et al., “Survey of hallucination in natural language generation,” *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.

[4] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009.

[5] V. Karpukhin, B. Oğuz, S. Min, et al., "Dense passage retrieval for open-domain question answering," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[6] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2009, pp. 758–759.

[7] D. Edge, H. Trinh, N. Cheng, et al., "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.

[8] L. Blecher, G. Cucurull, T. Scialom, and R. Stojnic, "Nougat: Neural optical understanding for academic documents," *arXiv preprint arXiv:2308.13418*, 2023.